How The IoT Is Revolutionising TELECOM

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IoT can propel both the top line and bottom line of telcos by introducing more operational efficiency and additional revenue streams

Given all the marketing push and hype around the Internet of Things (IoT), it is difficult to predict how business models will evolve over the next few years. For industry practitioners, this remains one of the major concerns. Usually, they ponder upon implementing the IoT and end up asking, "How can the IoT be used in my industry?"

Looking at the existing solutions is the most common approach to find an answer to this question. At one end, there are business-to-consumer (B2C) solutions, which are good to have but don't solve the real pain. At the other end, we see some B2B trends around particular industries where the IoT can play a crucial role. One such vertical is telecom.

For more than a decade, telecom has been the poster boy of high-technology industry. This is because of its fast expansion, highly driven by innovation around smartphones and the Internet. New services are being promoted and new geographies being captured by telcos. All this adds complexity to managing the work optimally, combined with the force of launching new services that add value to eliminate the competition.

Emergence of the IoT presents some interesting opportunities for telcos. Great news is that these opportunities can propel both the top line and bottom line, introducing more operational efficiency and additional revenue streams. Let us look at some of the available options.

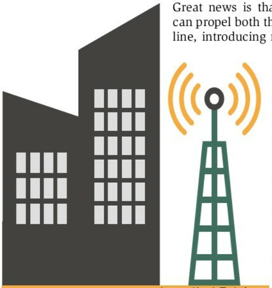

Image: Kaa IoT platform

Enhancing operational efficiency

Remote infrastructure accounts for a significant portion of telcos' operational cost. The IoT plays a significant role here in remote management and monitoring. Let us consider a mobile operator network, which contains many mobile cell towers that are spread across a large geographical area. For managing the remote equipment of telcos, protocols and standards are in place. However, there are issues like environmental protection, physical security and asset management.

Asset handling and monitoring. A remote cell tower site incorporates auxiliary equipment along with major telecommunication equipment in order to make things work. One such equipment is the generator for power backup, which ensures 24/7 network uptime. Energy meters, UPS and air-conditioning are other major assets. These come under the passive infrastructure. Predicting their failure in advance by monitoring their operational efficiency is an important part of remote management. The IoT makes it possible.

Physical security. In remote places, physical security of costly equipment is important. This requires an IoT-enabled intrusion detection system. In addition, a tracking system for resource consumption by pilferable consumables such as batteries and fuels is needed to ensure timely alerts in order to minimise losses.

Environmental protection. There is always the threat of environmental damage to remote sites. The IoT plays a major role in detecting conditions such as flood, smoke and bad weather. It also assists in providing control commands to either shut down the system to avoid irreparable damages or take any preventive measures.

Providing last-mile access to devices

When using the IoT to handle remote telco sites, it must be noted that the IoT depends on a reliable communication link. So it is crucial to consider something which doesn't suffer from downsides of the telco's terrestrial link failure.

Low-power radio. By playing the role of a service provider, telcos can tap into the machine-to-machine (M2M) ecosystem. M2M communication needs very low band-